WEEK 3 · 23–26 NOVEMBER 2026

Presenting, submitting and being evaluated

Twelve minutes to convince a room that your map can be trusted and used

Week 3 · Session 15 · practical: templates/presentation_template.md

LEARNING OBJECTIVES

What you will be able to do

- Structure a 12-minute technical presentation that a mixed audience can follow
- Present a map and a validation table so a non-specialist draws the right conclusion
- Answer the three questions every Earth observation project is asked
- Submit a complete, reproducible project folder
- Read the rubric the way the evaluators will read your work

1. The story arc

Problem → data → method → result → so what

HOW TECHNICAL TALKS FAIL

The audience decides in the first two minutes whether your result matters.

Start with the problem and the person who has it. A presentation that opens with the sensor specification has already lost half the room.

Twelve slides, twelve minutes

#	Slide	Time
1–3	Title · the problem · today's solution and its gap	2:00
4	Why hyperspectral — one spectrum, key wavelengths marked	1:00
5–6	Data · method (a diagram, not code)	2:20
7	The result — one large, readable map	1:30
8	Does it work? Validation, baseline, uncertainty	1:30
9–10	Limitations · from notebook to product	2:10
11–12	Impact and next steps · credits	1:00

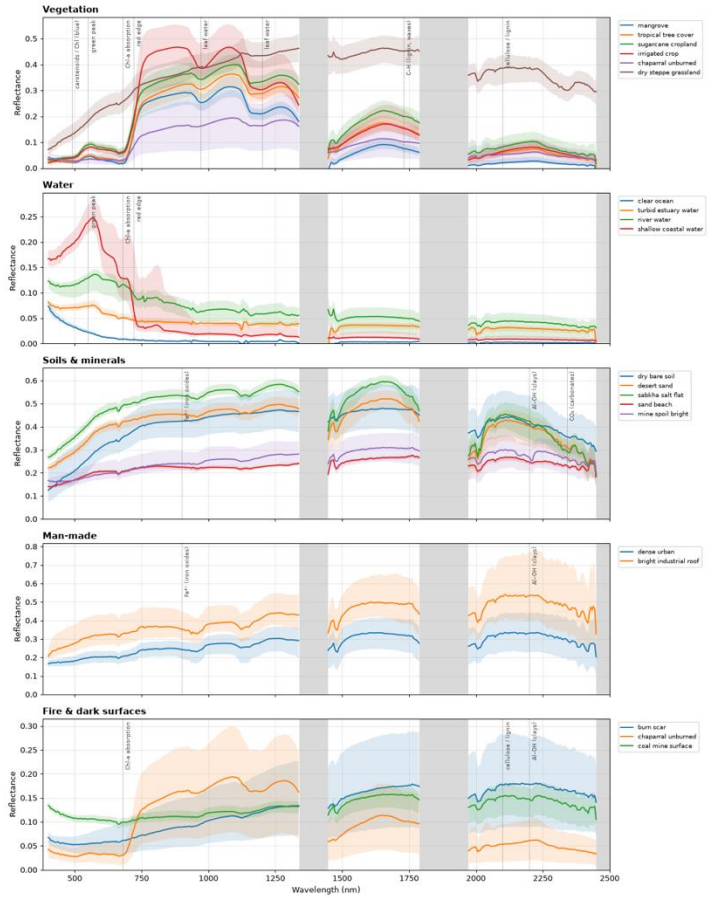
Titles that do the work for you

- Write the **takeaway** as the slide title, as a full sentence
- 'Mangrove loss concentrates near shrimp ponds' — not 'Results'
- One message per slide; if it needs two, it is two slides
- The audience reads the title before you speak — make that sentence count

SLIDE 4

The slide that convinces a sceptic

- Show a **spectrum**, not a specification table
- Mark the two or three wavelengths your method actually uses
- Say in one sentence what the feature is caused by
- This is where you earn the Creativity mark for using hyperspectral properly



Annotated spectra make the hyperspectral argument in seconds

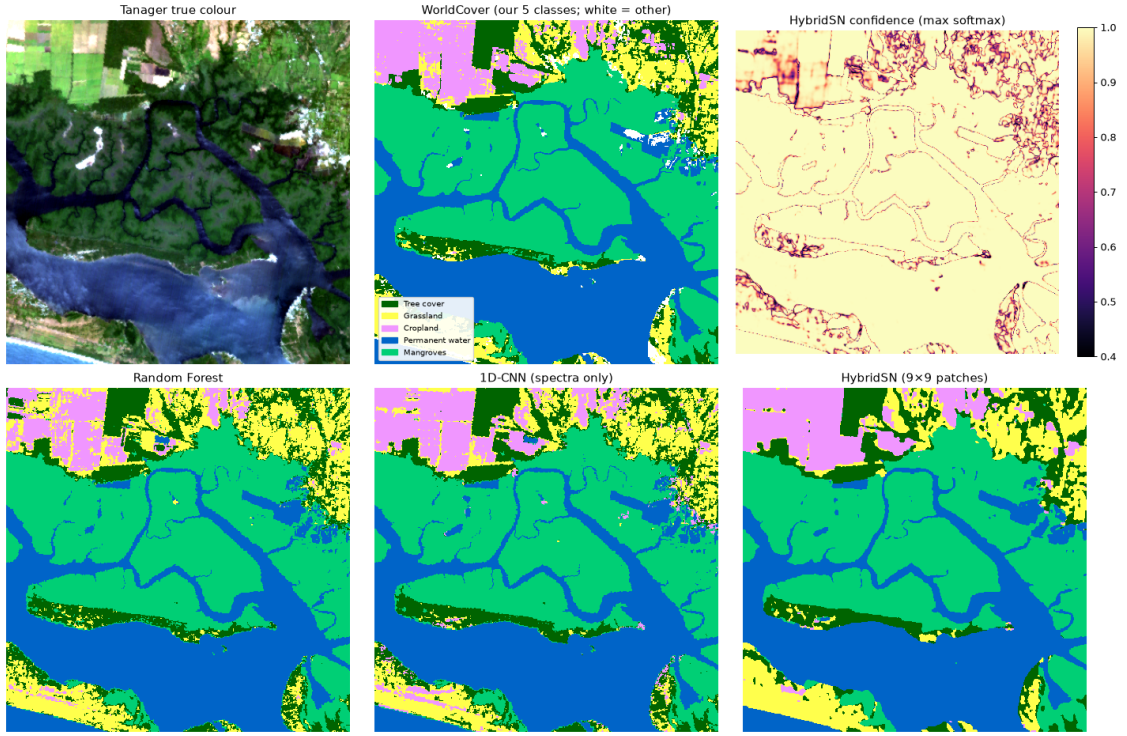
2. Showing results honestly

Maps and numbers a non-specialist can read

SLIDE 7

How to present a map

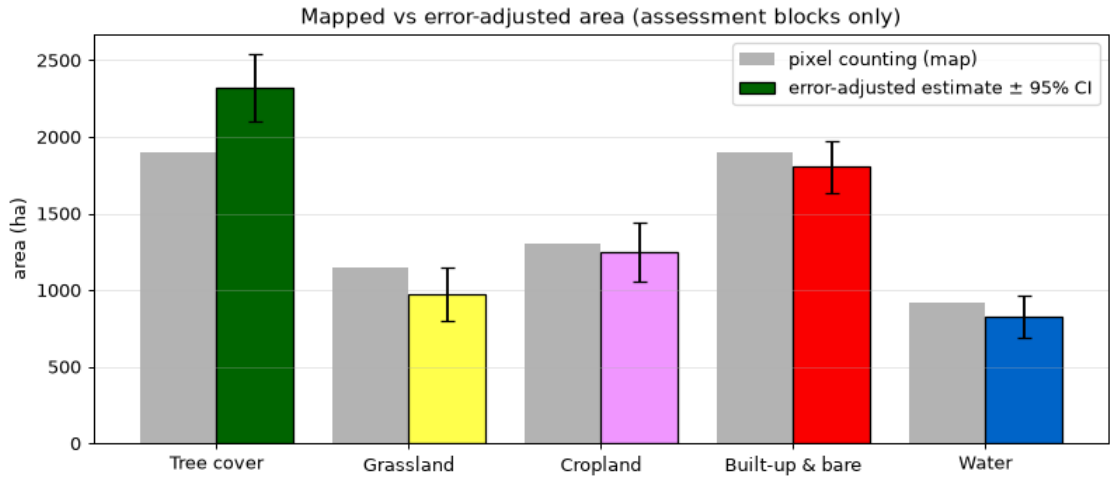
- Give context: where is this, how big, what am I looking at
- Legend in plain words — 'mangrove', not 'class 95'
- Keep the true-colour image beside the map for reference
- Same colours for the same classes on every slide
- Scale bar and north arrow; state the pixel size



Reference image, labels and model output side by side

How to present validation

- Lead with the **honest** number and its uncertainty: 'OA 0.60 ± 0.04'
- Put the per-class table on the slide; name the weakest class
- Show the **baseline** beside your model so the gain is visible
- Translate it: 'we miss one mangrove pixel in eight, at the water edge'
- Never show accuracy from a random pixel split



Area-adjusted accuracy and area, with 95 % confidence intervals

Limitations without undermining yourself

- Two or three real limitations, each with what you would do about it
- Frame as **scope**, not failure: 'validated for this estuary, not yet for Gulf mangroves'
- Name the data limits: single date, 30 m mixed pixels, 2021 reference labels
- A team that names its limitations is trusted more, not less

3. The three questions

Every team gets asked these

Prepare an answer for each, and rehearse it

- **How do you know it is right?** → sampling design, independent test data, the CI, the baseline
- **Why not just use Sentinel-2?** → name the feature and its width; show the resampled comparison
- **What would operational use take?** → repeat coverage, runtime, cost, who runs it, how often
  - If multispectral genuinely would do, say so and explain what hyperspectral adds beyond it
- Every team member should be able to answer all three, not just the presenter

4. Rehearse, submit, be evaluated

Wednesday and Thursday of the final week

TWO RUN-THROUGHS MINIMUM

Rehearsal discipline

- Time it twice, out loud, standing — silent reading is always faster than speaking
- Cut, do not speed up: at 12 minutes something must go, and it is never the validation
- Decide who speaks when, and how you hand over
- Fonts at least 18 pt; no screenshots of code; figures readable from the back
- Bring a PDF on a USB stick and a copy online; record a demo video as a fallback

TEMPLATES/SUBMISSION_CHECKLIST.MD

The submission folder

```
team_<name>_hyperspectral_project/
├── README.md               # problem, data, method, key results, how to run
├── project.ipynb           # executed, runs top to bottom
├── outputs/                # *.tif (COG) + *_metadata.json + figures/
├── validation/             # metrics tables, confusion matrix, plots
├── presentation.pdf
└── requirements.txt        # anything beyond the course requirements
```

No raw HDF5 files — scene IDs are enough for anyone to re-download them

EACH SCORED 1–4 · Σ (SCORE/4 × WEIGHT)

How the rubric scores you

Criterion	Weight	What a top mark looks like
Validity	30	Independent validation, CIs, baseline, failure cases explained
Relevance	20	Complete reproducible workflow with a path to operational use
Impact	20	Named user and decision, benefit quantified
Creativity	15	Uses spectral features only hyperspectral can resolve
Presentation	15	Clear story, readable figures, confident answers

After the course

- The top teams continue into the incubation phase, from proof of concept towards an MVP
- Keep the project **reproducible** — future you, or a new team member, needs to rerun it
- Keep data attributions intact if you publish or present the work elsewhere
- Consider releasing the notebook openly; it is a strong portfolio piece

PRACTICAL

Over to you

▶ `templates/presentation_template.md` · 150 min

- Draft the twelve slides, writing each title as a full-sentence takeaway
- Rehearse twice, timed and standing, then cut to fit
- Drill the three questions with a teammate answering at random
- Complete templates/submission_checklist.md and fix anything unticked

KEY TAKEAWAYS

Recap

- Open with the problem and the person who has it, not the sensor
- Titles are sentences; one message per slide
- Show a spectrum, show a readable map, show the honest number with its uncertainty
- Limitations framed as scope build trust
- Rehearse aloud, twice; submit a folder that runs without you

Next: Final evaluation and feedback — then the incubation pathway

References and further reading

templates/presentation_template.md · templates/submission_checklist.md · templates/evaluation_rubric.md (this course)

Schimel, J. (2012). Writing Science: How to Write Papers That Get Cited and Proposals That Get Funded. OUP.

Duarte, N. (2010). Resonate: Present Visual Stories that Transform Audiences. Wiley.

Tufte, E.R. (2001). The Visual Display of Quantitative Information, 2nd ed. Graphics Press.

Crameri, F. et al. (2020). The misuse of colour in science communication. Nature Communications 11, 5444.